Authorized to work in the U.S. without restrictions.

Peter Beigi, PhD Student

Autonomous Driving Researcher and Machine Learning Engineer

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Based in Santa Clara, CA & Washington, DC
Open to Relocate

PROFESSIONAL SUMMARY

Develops perception, prediction, and decision-making systems for autonomous driving. Experienced in building scalable ML pipelines using large-scale multimodal datasets across simulation and real-world environments.

RESEARCH EXPERIENCE

Research Intern
May 2026 - Aug 2026

University of Illinois Urbana-Champaign — Urbana-Champaign, IL

- Working with *Prof. Talebpour* on developing advanced approaches including foundation models, vision-language architectures, diffusion models, imitation learning, and reinforcement learning, leveraging large-scale driving data.
- Developed a utility-guided residual multi-agent RL framework for decentralized spatial negotiation and trajectory generation in weakly structured traffic *[paper]* *[code]*.
- Developed plan-conditioned diffusion world models for motion-as-communication negotiation at unprotected conflict points *[paper]* *[code]*.

Autonomous Vehicle & Machine Learning Intern
Sep 2022 - Present

FHWA Turner-Fairbank Research Center (Saxton Lab) — McLean, VA

- Leading development of trajectory generation models for ADAS (using physics-based, DL-based and RL-based frameworks), and integrating AV behavior models into large-scale microsimulation (SUMO) tools *[code]*.
- Developed and validated perception (sensor fusion, detection, segmentation) and planning (map generalization, localization) algorithms using LiDAR, radar, and video data for enhanced AV situational awareness and mixed-traffic simulation *[code]*.
- Built an end-to-end computer vision pipeline (Faster R-CNN, multi-camera MOT, BEV transformation, recursive state estimation, SIFT feature matching, and annotation/data processing) to extract trajectories and support motion prediction on 11,000+ vehicle-km of AV-human interaction data *[code]* *[Data]* **published by USDOT*.

Graduate Researcher
Sep 2021 - Present

The George Washington University — Washington, DC

- Engineered model observability and traceability by developing Logsim DAGs to capture closed-loop metrics, improving experiment reproducibility for the autonomous stack.
- Led creation of an integrated simulation-AI pipeline combining SUMO, CARLA, and Python APIs for closed-loop decision-making experiments using real-time data (trajectory) streaming using twelve 5G-connected cameras.

WORK EXPERIENCE

Autonomous Vehicle Researcher Intern
Sep 2024 - Dec 2024

moment.ai (startup under **State Farm Insurance**) — Washington, DC

- Developed multi-modal deep learning time-series models (LSTM/GRU) integrating smartphone and AV sensor data to detect driver behavior and health incident risks, outperforming traditional methods by 9%.
- Developed a real-time NVIDIA Jetson/mobile driver monitoring system using AWS IoT pipelines and Alexa integration for in-vehicle alerts and control.

SELECTED PUBLICATIONS

h-index: 6 / citation: 230

"Utility-Guided Residual Multi-Agent Reinforcement Learning for Closed-Loop Interactive Driving" *[code]*	IEEE T-ITS (under review)
"Toward Autonomous Driving in Weak-Lane Freeway Traffic: A Naturalistic Trajectory Dataset and Comparative Evaluation of Interaction Models" *[code]*	TR-C (under review)
"Empirically Calibrated Differential Games for Simulating Driver Interactions" *[arXiv]* *[data]*	IEEE T-ITS (under review); Oral at INFORMS
"Complete Street Design informed by Multi-Agent Simulation" *[arXiv]*	TRR (under review)
"Motion as Communication: Plan-Conditioned Diffusion World Models for Negotiation at Unprotected Conflict Points" *[pdf]* *[code]*[1]	IROS 2026
"Bridging Perception, Planning and Control: A System-Level Validation of a Microscopic Acceleration Model for Longitudinal Autonomous Driving" *[pdf]* *[code]*[2]	IROS 2026
"Deep Reinforcement Learning Under Partial Observability: Co-Simulation and Penetration-Aware Policy Learning for Traffic Control" *[code]*	IEEE VTC Fall (2026)
"Diffusion Process-Based Model for Network Trajectory Propagation" *[pdf]*	IEEE T-ITS (2026)
"Percolation-Based Cloud Computing Framework for Network Reliability Assessment" *[pdf]*	IEEE ICMI (2025)
"A Hybrid Genetic Algorithm and Reinforcement Learning Approach for Vulnerable Road User Modeling" *[code]*	TRR (2026)
"A Deep Reinforcement Learning Control to Explore Impact of V2X Connectivity on Intersection Performance" *[pdf]* *[code]*	TRR (2025)
"Third Generation Simulation Dataset" *[pdf]* *[code]* *[data]*	TRR (2024)

SERVICE

Session Chair, IEEE VTC

Reviewer: NeurIPS; IEEE/RSJ IROS; IEEE/CVF WACV; IEEE T-ITS; IEEE VTC; IEEE IV; Transportation Research Part C; and many more

TECHNICAL SKILLS

Programming: Python, ROS, MySQL, Git

Libraries: PyTorch, Diffusers, Transformers,OpenCV, RLlib, Gymnasium, Scikit-learn, Keras, Ultralytics

Cloud Computing Platform: Azure ML

Datasets: Waymo Motion, Waymo Perception, nuScenes, Argoverse 2, TGSIM, HighD, INTERACTION

Tools: CARLA, Unity, Power BI, Tableau, SUMO, VISSIM, Synchro, SIDRA

EDUCATION

The George Washington University
Sep 2021 - Exp: May 2027
PhD in Transportation Engineering (AV Systems & AI) - GPA: 3.91/4.00

Sharif University of Technology
Sep 2016 - Feb 2021
Bachelor of Science in Civil Engineering - GPA: 3.80/4.00

WORK AUTHORIZATION

No sponsorship or visa-based work authorization (CPT, OPT, H-1B, or otherwise) required.

Available to start immediately. Available with 3 days' notice for positions in the DC and Bay Area. Available with 1 week's notice for positions in other locations.

1. Lightning Presentation and Poster at RoBoWoMo Workshop.
2. Spotlight Presentation and Poster at BEMHAT Workshop.